

Pediatric emergencies in the rural ED: a physician education guide for the first 24 hours

Rural emergency departments manage approximately 20% of all pediatric emergency visits in the United States, yet rural physicians encounter critically ill children infrequently — making systematic preparation essential for survival. This guide synthesizes current evidence-based guidelines from the AAP, ACEP, PECARN, the Surviving Sepsis Campaign, AHA/PALS, and protocols from leading children's hospitals (CHOP, Texas Children's, Cincinnati Children's, Boston Children's, Nationwide Children's) into a practical clinical decision-making tool. It covers the first 24 hours of stabilization before transfer, with weight-based dosing tables, severity scoring systems, and rural-specific considerations for every major pediatric emergency. The guiding principle throughout: **stabilize, treat what you can, and transfer early** — because in rural pediatric emergency care, the most dangerous error is under-recognizing severity.

PART I: GENERAL PRINCIPLES AND REFERENCE TABLES

The rural ED faces unique challenges that demand proactive solutions

Rural emergency departments see fewer pediatric critical cases, which means lower provider familiarity with pediatric-specific physiology, equipment, and dosing. (PubMed Central) **Children are not small adults** — they compensate for shock with tachycardia and increased systemic vascular resistance until they abruptly decompensate, their airways are anatomically different, and nearly all medication dosing is weight-based.

Core strategies for rural ED preparedness include maintaining pre-calculated weight-based dosing sheets (or Broselow tape at every bedside), running quarterly pediatric simulation drills, establishing telemedicine partnerships with regional children's hospitals, and stocking a dedicated pediatric resuscitation cart with age-appropriate equipment. The AAP/ACEP Joint Policy Statement on Equipment for Emergency Departments Caring for Children provides the definitive checklist.

Key principles:

- **Measure or estimate weight immediately** — every intervention depends on it
- **Use cognitive aids** — laminated protocols, dosing apps, Broselow tape — without shame
- **Activate transfer early** — do not wait for clinical deterioration
- **Telemedicine consultation** changes outcomes — programs like Project ECHO provide real-time specialist support (Agency for Healthcare Research...)
- **Family presence** during resuscitation is supported by evidence and should be accommodated when safe

Pediatric vital signs by age

Age	Heart Rate (bpm)	Respiratory Rate	SBP (mmHg)	Hypotension (SBP)	Weight (kg)
Premature	120–170	40–70	55–75	<55	1–2.5
Newborn (0–1 mo)	100–170	30–60	60–76	<60	3–4.5
1–6 months	100–160	30–55	70–90	<70	4–8
6–12 months	90–150	24–40	72–100	<70	8–10
1–3 years	80–140	22–34	80–100	<70 + (age×2)	10–14
3–6 years	75–120	20–28	85–110	<70 + (age×2)	14–20
6–12 years	70–110	16–24	90–120	<70 + (age×2)	20–40
12–18 years	60–100	12–20	100–130	<90	40–70

Minimum urine output targets: Infants 1–2 mL/kg/hr; children 1 mL/kg/hr; adolescents 0.5 mL/kg/hr.

Weight estimation methods

Broselow tape remains the fastest and most validated tool. It is measured from head to heel in a supine child and provides color-coded zones with pre-calculated drug doses, equipment sizes, and fluid volumes. [Springer](#)

[PubMed Central](#)

Age-based formulas when tape is unavailable:

- Infants 1–12 months: $(\text{age in months} + 9) \div 2$
- 1–5 years: $2 \times (\text{age in years} + 5)$
- 6–12 years: $3 \times \text{age} + 7$ [PubMed Central](#)

For obese children, use **ideal body weight** for most medication dosing (exceptions: succinylcholine and certain antibiotics use actual weight). Actual weight should be measured whenever possible.

PART II: RESPIRATORY EMERGENCIES

Acute asthma and status asthmaticus demand aggressive, escalating treatment

Asthma exacerbations are classified by the **Pediatric Asthma Score (PAS)**: mild (5–7), moderate (8–11), severe (12–15). [PubMed Central](#) The core first-hour actions are identical regardless of setting: **oxygen, inhaled beta-**

agonists, systemic corticosteroids, and ipratropium for moderate-severe presentations.

First-hour critical actions:

Intervention	Dose	Notes
Albuterol (mild-moderate)	MDI 4–8 puffs q20min ×3	MDI + spacer equals nebulizer efficacy
Albuterol (severe/continuous)	0.5 mg/kg/hr nebulized (max 15–20 mg/hr)	CHOP protocol: 7.5 mg/hr (<10 kg), 11.25 mg/hr (10–20 kg), 15 mg/hr (>20 kg) (Children's Hospital of Philadelp...)
Ipratropium bromide	250 mcg (<20 kg) or 500 mcg (≥20 kg) nebulized q20min ×3	Add to first 3 albuterol treatments in moderate-severe
Dexamethasone	0.6 mg/kg PO/IM/IV (max 16 mg), single dose	Preferred over prednisone course; 2-dose regimen (day 1 and day 2) equivalent to 5-day prednisone
IV magnesium sulfate	25–75 mg/kg IV (typically 50 mg/kg, max 2 g) over 20 min	For severe exacerbation unresponsive to initial therapy; monitor for hypotension (Texas Children's)
IV terbutaline	Load: 10 mcg/kg over 10 min, then 0.1–10 mcg/kg/min	Requires continuous cardiac monitoring; risk of hypokalemia, tachycardia
Subcutaneous epinephrine	0.01 mg/kg of 1:1,000 (max 0.3 mg)	For impending arrest without IV access

Ventilator management in intubated asthma (avoid intubation if at all possible — peri-intubation cardiac arrest risk is high): Use ketamine 1–2 mg/kg IV for induction (bronchodilatory properties), (NCBI) rocuronium 1 mg/kg for paralysis. (Learnpicu) Set ventilator to **low rate (10–12/min), long expiratory time (I:E ratio 1:3 to 1:4), tidal volume 6–8 mL/kg, PEEP 0–5 cmH₂O, and tolerate permissive hypercapnia** (target pH ≥7.15–7.20). The single greatest pitfall is aggressive bagging after intubation, which causes air trapping, tension physiology, and arrest. If sudden hemodynamic collapse occurs post-intubation, **disconnect the ventilator and compress the chest** to release trapped air.

Transfer criteria: Failure to improve after continuous albuterol + steroids + magnesium; need for terbutaline infusion; (Ucsfbenioffchildrens) intubation; hypercarbia on blood gas.

Bronchiolitis management is primarily supportive — less is more

Per the **AAP 2014 Clinical Practice Guideline**, bronchiolitis management centers on supportive care: (AAFP) gentle nasal suctioning, supplemental oxygen if SpO₂ persistently ≤90%, and hydration support. **Routine use of**

bronchodilators, corticosteroids, and antibiotics is not recommended. (AAFP) This evidence-free tradition of treating bronchiolitis like asthma persists in practice and should be actively resisted.

High-flow nasal cannula (HFNC) has transformed bronchiolitis management. Start at **2 L/kg/min**

(Lippincott Williams & Wilkins) (max 30–50 L/min depending on device), titrate FiO_2 to maintain $\text{SpO}_2 \geq 90\text{--}92\%$.

HFNC generates positive airway pressure ($\sim 4 \text{ cmH}_2\text{O}$ at 2 L/kg/min), reduces work of breathing, and is safe in community hospitals without PICU capability. **Escalate to CPAP/BiPAP or intubation** if FiO_2 requirement exceeds 50–60% on maximum flow, work of breathing worsens, apnea occurs, or mental status deteriorates.

Transfer criteria: Apnea episodes, need for CPAP/BiPAP/intubation, $\text{FiO}_2 > 40\text{--}50\%$ on HFNC, progressive respiratory failure, or age <12 weeks with risk factors.

Croup responds dramatically to dexamethasone and racemic epinephrine

The **Westley Croup Score** (0–17) grades severity:

Component	0	1	2	3	4–5
Stridor	None	With agitation	At rest	—	—
Retractions	None	Mild	Moderate	Severe	—
Air entry	Normal	Mildly ↓	Markedly ↓	—	—
Cyanosis	None	—	—	—	With agitation (4) / At rest (5)
Consciousness	Normal	—	—	—	Disoriented (5) (Medscape)

Mild ≤ 2 (85% of cases); moderate 3–5; severe 6–11; impending respiratory failure ≥ 12 . (Pediatric Education)

(Medscape)

All severities receive dexamethasone 0.6 mg/kg PO (max 16 mg), single dose. (American Academy of Family ...)

For moderate-severe (stridor at rest), add **nebulized racemic epinephrine 0.05 mL/kg of 2.25% (max 0.5 mL)** or **L-epinephrine 0.5 mL/kg of 1:1,000 (max 5 mL)** (American Academy of Family ...)

— L-epinephrine is equally effective and universally available. (Aspo) **Observe minimum 2–4 hours post-epinephrine** for rebound. If no improvement after ≥ 2 epinephrine doses, consider alternative diagnoses: epiglottitis (no barking cough, drooling, toxic), bacterial tracheitis (toxic deterioration after viral prodrome), or retropharyngeal abscess. (NCBI)

Foreign body aspiration requires recognition and transfer for bronchoscopy

Suspect in any child with acute choking, unilateral wheeze, or asymmetric breath sounds — especially toddlers aged 1–3 years. (PubMed Central) CXR is **normal in >50%** of cases. (Medscape) Inspiratory/expiratory films may show hyperinflation of the affected side. **For acute complete obstruction:** infants <1 year get 5 back blows + 5 chest thrusts; children >1 year get abdominal thrusts (Heimlich). For stable patients, keep calm, provide supplemental O_2 , maintain NPO, and **arrange urgent transfer to a center with pediatric rigid bronchoscopy**.

Pneumonia antibiotic selection depends on age

Age	Outpatient	Inpatient
0–28 days	Admit all	Ampicillin 50 mg/kg IV q8h + gentamicin 4–5 mg/kg IV q24h
1–3 months	Admit most	Ceftriaxone 50–75 mg/kg IV daily (or ampicillin IV)
3 mo – 5 yr	Amoxicillin 90 mg/kg/day PO BID	Ampicillin 150–200 mg/kg/day IV q6h (Texas Children's)
>5 years	Amoxicillin + azithromycin 10 mg/kg day 1, then 5 mg/kg days 2–5	Ampicillin or ceftriaxone + azithromycin

For complicated pneumonia with **pleural effusion/empyema**: chest tube + fibrinolysis (tPA 4 mg in 40 mL NS) or VATS; antibiotics broadened to ceftriaxone + vancomycin (Texas Children's) pending cultures.

Respiratory failure: the RSI protocol every rural physician must know

Recognition: Age-adjusted tachypnea, nasal flaring, retractions, grunting, head bobbing → altered mental status and cyanosis (late/ominous). Trial **CPAP (5–8 cmH₂O)** or **BiPAP (IPAP 10–16, EPAP 5–8)** before intubation when possible. (JEMS)

RSI medication table:

Step	Agent	Dose	Notes
Pretreatment	Atropine	0.02 mg/kg IV (min 0.1 mg, max 0.5 mg)	For age <1 yr or succinylcholine use (Medscape)
Induction	Ketamine	1–2 mg/kg IV	Preferred for asthma, shock (hemodynamically supportive) (Learnpicu)
Induction	Etomidate	0.3 mg/kg IV	Hemodynamically neutral; avoid in sepsis (Learnpicu)
Induction	Propofol	1–2 mg/kg IV	Causes hypotension; use cautiously (Learnpicu)
Paralytic	Rocuronium	1 mg/kg IV	Preferred; onset 45–60 sec; (PubMed Central) reversible with sugammadex
Paralytic	Succinylcholine	2 mg/kg IV (infants), 1–1.5 mg/kg (children)	Onset 30 sec; contraindicated in hyperkalemia, burns >24 hr, neuromuscular disease (LITFL)

ETT sizing: Cuffed ETT = (age/4) + 3.5; uncuffed = (age/4) + 4. **ETT depth at lip** = (age/2) + 12 cm, or 3 × ETT internal diameter. Always have half-size above and below available. Confirm placement with **end-tidal CO₂** (gold standard), auscultation, and CXR.

Equipment sizing by age:

Age	ETT (cuffed)	Blade	LMA	Suction (Fr)
Neonate	3.0–3.5	Miller 0–1	1	8
6 months	3.5	Miller 1	1.5	8
1 year	3.5–4.0	Miller 1	1.5–2	8–10
4 years	4.5	Miller/Mac 2	2–2.5	10–12
8 years	5.5	Mac 2–3	2.5–3	12
12 years	6.5	Mac 3	3–4	14

PART III: SEPSIS AND SEPTIC SHOCK

The Hour-1 Bundle saves lives — execute it without delay

Pediatric sepsis kills through a cascade of inadequate perfusion, organ dysfunction, and metabolic failure. Medscape The **Surviving Sepsis Campaign 2020 Pediatric Guidelines** (Weiss et al., *Pediatr Crit Care Med* 2020) provide the definitive framework. The **2024 Phoenix Sepsis Score** (Schlapbach et al., *JAMA* 2024) has replaced older SIRS-based definitions: a Phoenix Score ≥2 in a child with suspected infection defines sepsis; add cardiovascular dysfunction for septic shock. Medscape

Hour-1 Bundle — execute simultaneously, not sequentially:

- 1. **Blood cultures** (before antibiotics, unless it would delay them)
- 2. **Broad-spectrum antibiotics** within 1 hour for septic shock (strong recommendation)
- 3. **Isotonic crystalloid boluses** 10–20 mL/kg over 5–10 min; reassess after each bolus
- 4. **Measure lactate**
- 5. **Reassess perfusion** after each bolus: capillary refill, heart rate, mental status, urine output

Fluid resuscitation: Give up to **40–60 mL/kg** in the first hour in facilities with PICU; up to **40 mL/kg** if no PICU available (SSC weak recommendation). Use **balanced crystalloids (LR preferred)** over NS. After each

10–20 mL/kg bolus, examine for fluid overload: hepatomegaly, crackles, worsening respiratory distress. **Never use hydroxyethyl starch** (strong recommendation against).

Cold shock versus warm shock determines vasopressor choice

Cold shock (~70% of pediatric septic shock): prolonged capillary refill >3 seconds, cool/mottled extremities, narrow pulse pressure, weak pulses ([SaveaLife.com](#)) → treat with **epinephrine 0.1–1 mcg/kg/min**.

Warm shock: flash capillary refill (<1 sec), bounding pulses, wide pulse pressure, warm flushed skin ([SaveaLife.com](#)) → treat with **norepinephrine 0.1–2 mcg/kg/min**.

Vasopressor	Dose	Primary Indication
Epinephrine	0.1–1 mcg/kg/min	Cold shock (first-line)
Norepinephrine	0.1–2 mcg/kg/min	Warm shock (first-line)
Dopamine	5–20 mcg/kg/min	Only if epi/norepi unavailable
Vasopressin	0.0003–0.002 units/kg/min	Adjunct for catecholamine-resistant shock

Critical rural pearl: Vasopressors can be initiated through a **peripheral IV in dilute concentration** if central access is unavailable — do not delay pressors waiting for a central line.

Empiric antibiotic selection by age and source

Age	Empiric Regimen	Key Coverage
0–28 days	Ampicillin + gentamicin	GBS, E. coli, Listeria (ceftriaxone contraindicated in neonates)
1–3 months	Ampicillin + ceftriaxone	Residual neonatal pathogens + community organisms
3 months–18 yr	Ceftriaxone ± vancomycin	Add vancomycin for MRSA concern or indwelling devices
Abdominal source	Add metronidazole	Anaerobic coverage
HSV concern	Add acyclovir 20 mg/kg IV q8h	Neonates with vesicles, seizures, or elevated LFTs
Neutropenic	Cefepime or meropenem	Anti-pseudomonal coverage essential

Stress-dose hydrocortisone (1–2 mg/kg IV bolus, then 50 mg/m²/day divided q6–8h): Consider for catecholamine-resistant shock, suspected adrenal insufficiency, or chronic steroid use. Do not use etomidate for intubation in septic shock (adrenal suppression). ([Learnpicu](#))

PART IV: PEDIATRIC SHOCK — ALL TYPES

Recognizing compensated shock before decompensation is the rural physician's greatest skill

Children maintain blood pressure through tachycardia and vasoconstriction until they cannot — then they crash precipitously. (SaveaLife.com) **Tachycardia is the earliest sign of shock. Hypotension is a late, ominous finding.**

Hypotension thresholds (systolic):

- Neonates: <60 mmHg
- 1–12 months: <70 mmHg
- 1–10 years: $<70 + (\text{age} \times 2)$ mmHg (pedscales)
- **10 years:** <90 mmHg

Hypovolemic shock is the most common type in children

Causes: gastroenteritis (most common worldwide), hemorrhage, burns. (SaveaLife.com) Manage with **20 mL/kg isotonic crystalloid boluses** (NS or LR), repeating up to 60 mL/kg. (SaveaLife.com) For hemorrhagic shock unresponsive to 2–3 crystalloid boluses, transfuse **pRBCs 10 mL/kg** (Aclscertification) (O-negative for females, type-O for males if type-specific unavailable). (Aclscertification)

Cardiogenic shock requires a fundamentally different approach

Recognition clues that distinguish cardiogenic from other shock types: hepatomegaly, jugular venous distension, pulmonary crackles/edema, gallop rhythm, and labored breathing (versus quiet tachypnea in hypovolemic shock). (SaveaLife.com) Point-of-care ultrasound is transformative here.

AVOID aggressive fluid resuscitation. Give only cautious 5–10 mL/kg boluses over 10–20 minutes if needed. Primary therapy is **inotropic support:** (SaveaLife.com)

- **Dobutamine 5–20 mcg/kg/min** (easiest to start — no loading dose needed)
- **Milrinone 0.25–0.75 mcg/kg/min** (preferred long-term; loading dose 50 mcg/kg over 10–60 min optional)

Causes in children: myocarditis, cardiomyopathy, congenital heart disease, dysrhythmias, Kawasaki disease. **Transfer to a center with ECMO capability.**

Obstructive shock requires immediate cause-specific intervention

Tension pneumothorax: Clinical diagnosis (SaveaLife.com) — do not wait for CXR. Needle decompression at 2nd intercostal space, midclavicular line (14–18 gauge) followed by chest tube.

Cardiac tamponade: Beck's triad (muffled heart sounds, JVD, hypotension). (SaveaLife.com) Fluid boluses temporize while preparing for emergency pericardiocentesis.

Ductal-dependent congenital heart lesions (neonates presenting at 1–2 weeks as ductus closes): Start **prostaglandin E1 (alprostadil) 0.05–0.1 mcg/kg/min IV** (University of Iowa Health Care) (can decrease to 0.01–0.05 once ductus opens). **Have intubation equipment ready** — apnea occurs in ~40%. (PubMed Central) Side effects include hypotension, fever, and bradycardia. (Q Wave)

PART V: ANAPHYLAXIS

Epinephrine is the only first-line treatment — everything else is adjunctive

Anaphylaxis is diagnosed when illness involves **2 or more organ systems** after exposure to a likely allergen (NIAID/WAO criteria). Up to 80–90% of children have cutaneous involvement, but **absence of skin findings does not rule out anaphylaxis**.

Epinephrine IM dosing (anterolateral thigh):

Weight	Dose	Delivery
<7.5 kg	0.01 mg/kg	Draw from 1 mg/mL vial
7.5–25 kg	0.15 mg	EpiPen Jr auto-injector
>25–30 kg	0.3 mg	EpiPen auto-injector
>30 kg / adolescent	0.3–0.5 mg	Max 0.5 mg per dose

Repeat every 5–15 minutes as needed. (American Academy of Pediatrics) There are **no absolute contraindications** to epinephrine in anaphylaxis.

Refractory anaphylaxis (persistent symptoms after ≥ 2 IM doses): Start **IV epinephrine infusion** (Jacionline) **0.1–1 mcg/kg/min**, aggressive IV fluid boluses (20 mL/kg NS, may need 60+ mL/kg), and consider vasopressin or norepinephrine for persistent vasodilatory shock.

Second-line treatments (never delay epinephrine for these):

- Diphenhydramine 1 mg/kg IV/IM (max 50 mg) — for urticaria only
- Famotidine 0.25 mg/kg IV (max 20 mg) — H2 blocker (ranitidine withdrawn)
- Albuterol 2.5–5 mg nebulized — for bronchospasm
- Methylprednisolone 1–2 mg/kg IV (max 125 mg) — evidence for preventing biphasic reactions is weak
- **Glucagon for beta-blocked patients:** 20–30 mcg/kg IV (max 1 mg)

Observe minimum 4–6 hours after last epinephrine dose. Extend to **8–24 hours for severe reactions** requiring multiple epinephrine doses or causing hypotension. **Biphasic reactions occur in 1–20%** of cases, typically within 8–12 hours. Discharge with auto-injector prescription and allergy referral.

PART VI: DEHYDRATION AND FLUID/ELECTROLYTE EMERGENCIES

Clinical dehydration assessment guides the rehydration strategy

The **Clinical Dehydration Scale (CDS)** by Friedman is the only prospectively validated tool (scores general appearance, eyes, mucous membranes, tears on a 0–8 scale: 0 = none; 1–4 = some; 5–8 = moderate/severe dehydration). The **Gorelick 4-point scale** (capillary refill >2 sec, absent tears, dry mucous membranes, ill appearance) is also practical: ≥ 2 signs suggests $\geq 5\%$ dehydration.

Severity	% Dehydration	Key Signs
Mild	3–5%	Slightly dry mucous membranes, normal cap refill, mildly decreased urine
Moderate	6–9%	Sunken eyes, decreased tears, delayed cap refill (2–3 sec), tachycardia
Severe	$\geq 10\%$	Deeply sunken eyes, absent tears, tenting skin, cap refill >3 sec, hypotension, lethargy

Oral rehydration therapy (ORT) is first-line for mild-moderate dehydration. (NCBI) Give **50 mL/kg (mild) to 100 mL/kg (moderate) over 4 hours** in small frequent sips. (Merck Manual) A single dose of **ondansetron 0.15 mg/kg PO (max 4 mg)** reduces vomiting (PubMed Central) from 35% to 14% and halves the need for IV rehydration (Freedman et al., NEJM 2006). Nasogastric rehydration is an effective alternative when oral intake is insufficient.

IV fluid resuscitation uses isotonic crystalloid (NS or LR). Bolus 20 mL/kg over 20 minutes for dehydration; 10–20 mL/kg rapidly for shock. **Maintenance fluids use the Holliday-Segar (4-2-1) method:** 4 mL/kg/hr for first 10 kg + 2 mL/kg/hr for 10–20 kg + 1 mL/kg/hr for each kg >20. The **AAP 2018 guideline strongly recommends isotonic maintenance fluids (D5 NS)** rather than hypotonic solutions, (Medscape) as iatrogenic hyponatremia from hypotonic fluids causes 15–30% of hospitalized children to become hyponatremic.

DKA in children kills through cerebral edema — the protocol must be followed exactly

DKA definition: glucose >200 mg/dL + pH <7.3 or bicarb <15 + ketonemia/ketonuria. Severity: mild (pH 7.2–7.3, bicarb 10–15), moderate (pH 7.1–7.2, bicarb 5–10), **severe (pH <7.1, bicarb <5)**. (PubMed Central)

Step-by-step DKA protocol:

1. Initial fluid resuscitation (first hour): 10–20 mL/kg 0.9% NS bolus. (Children's Hospital of Philadelp...) **Do not exceed 40 mL/kg** in initial boluses unless frank shock. Establish 2 peripheral IVs. Obtain VBG, BMP, Mg, Phos, glucose, and beta-hydroxybutyrate. (chop)

2. Insulin infusion: Start **regular insulin at 0.05–0.1 units/kg/hr** AFTER initial bolus. (chop) **NEVER bolus insulin in pediatric DKA** (increases cerebral edema risk). Goal glucose decrease: **50–100 mg/dL/hr**.

(Children's Hospital of Philadelp...) When glucose reaches 250–300 mg/dL, **add dextrose (D5 or D10) to IV fluids — do NOT stop insulin**. Continue insulin until the anion gap closes (bicarb >15, BOHB <2 mmol/L). If glucose drops too quickly, increase dextrose concentration rather than decreasing insulin.

3. Ongoing fluids: Replace deficit over **24–48 hours** (not faster). Do not exceed **1.5–2× maintenance rate**.

(Texas Children's) Use the two-bag system (one bag with D10 NS, one with plain NS) to allow rapid dextrose titration based on hourly glucose.

4. Potassium replacement (total body K^+ is always depleted despite normal/high initial levels):

Serum K^+	KCl in IV fluids	Action
<3.5 mEq/L	40 mEq/L	HOLD insulin until K^+ corrected
3.5–5.4 mEq/L	20–40 mEq/L	Replace and continue insulin
5.5–6.0 mEq/L	10 mEq/L	Monitor closely
>6.0 mEq/L	None	Recheck in 1 hour

Replace half of potassium as KPhos to prevent hypophosphatemia. **Do NOT give bicarbonate** unless pH <6.9 with hemodynamic compromise (independent risk factor for cerebral edema).

5. Cerebral edema occurs in **0.5–1% of DKA episodes** (Thesgem) with 20–25% mortality. Risk factors include new-onset diabetes, age <5 years, severe DKA, excessive fluids, rapid glucose correction, and bicarbonate use.

Recognize by: headache, altered mental status, bradycardia, hypertension, pupillary changes. **Treat immediately without waiting for CT:**

- **Mannitol 0.5–1 g/kg IV over 20 min OR 3% hypertonic saline 2.5–5 mL/kg IV over 10–15 min**
- Elevate head of bed 30 degrees
- Reduce IV fluid rate by one-third
- Monitor corrected sodium (should RISE during treatment — failure to rise is a red flag)

Monitoring: Glucose hourly, BMP/VBG q2h initially, neurological assessment hourly, vitals q15 min initially then hourly.

The PECARN DKA FLUID Trial (Kuppermann et al., NEJM 2018) — a landmark 13-center RCT of 1,255 children — found that neither the rate nor sodium content of IV fluids significantly influenced neurological outcomes. However, current ISPAD guidelines still recommend cautious rehydration pending further data.

Sodium and glucose emergencies require precise correction rates

Symptomatic hyponatremia (seizures/altered mental status): Give **3% hypertonic saline 2–5 mL/kg IV over 10–20 minutes**. Goal: raise Na by 4–6 mEq/L acutely. Maximum correction: **6–8 mEq/L per 24 hours** (risk of osmotic demyelination with overcorrection).

Hypernatremia: Correct **no faster than 0.5 mEq/L/hr or 10–12 mEq/L per 24 hours**. Rapid correction causes cerebral edema. Restore intravascular volume with isotonic saline first, then correct free water deficit slowly over 48–72 hours.

Hypoglycemia treatment by age:

Age	Solution	Dose
Neonates	D10W	2 mL/kg IV
Infants	D10W	2–5 mL/kg IV
Children	D25W	2–4 mL/kg IV
Adolescents	D50W	1–2 mL/kg IV (max 25 g)
No IV access (any age)	Glucagon IM	0.5 mg (<25 kg), 1 mg (≥25 kg)

Never use D50W in neonates or young infants — it is hyperosmolar and causes intracranial hemorrhage and phlebitis.

PART VII: NEUROLOGICAL EMERGENCIES

PECARN rules prevent unnecessary CT scans while catching dangerous injuries

The **PECARN head injury prediction rules** (Kuppermann et al., Lancet 2009; 42,412 children) are the most validated clinical decision tool in pediatric emergency medicine.

Children <2 years — very low risk if ALL absent:

1. Altered mental status
2. Non-frontal scalp hematoma
3. Loss of consciousness ≥5 seconds

4. Severe mechanism of injury
5. Palpable skull fracture
6. Not acting normally per parents

Children ≥ 2 years — very low risk if ALL absent:

1. Altered mental status
2. Loss of consciousness
3. Vomiting
4. Severe mechanism
5. Signs of basilar skull fracture
6. Severe headache

Sensitivity 100% (age < 2) and 96.8% (age ≥ 2) for clinically important TBI. If all criteria absent $\rightarrow$ discharge with return precautions. If highest-risk features present (GCS < 15 , palpable skull fracture, signs of basilar fracture) $\rightarrow$ CT. If intermediate risk $\rightarrow$ observation for 4–6 hours with shared decision-making is reasonable.

Pediatric GCS — modified verbal scale for preverbal children (< 2 years):

Score	Standard Verbal	Preverbal Modification
5	Oriented	Coos, babbles, smiles appropriately
4	Confused	Cries but consolable
3	Inappropriate words	Persistently irritable
2	Incomprehensible	Restless, agitated, moaning
1	None	None

ICP management in the rural ED before transfer (for GCS ≤ 8 or signs of herniation):

- **Prevent secondary brain injury** (most critical): avoid hypotension (maintain age-appropriate SBP) and hypoxia (SpO₂ $\geq 95\%$)
- Head of bed elevated 30 degrees, midline head position
- Intubate if GCS ≤ 8 , target PaCO₂ 35–42 mmHg (normocarbida)
- **For herniation signs:** 3% hypertonic saline 2–5 mL/kg IV over 10–20 min (preferred in pediatrics) or mannitol 0.5–1 g/kg IV

- Hyperventilate only as a **temporizing bridge** (target PaCO₂ ~35; avoid aggressive hyperventilation <30)
- Seizure prophylaxis: levetiracetam 60 mg/kg IV or fosphenytoin 20 mg PE/kg IV

Status epilepticus: time is brain — follow the benzodiazepine ladder

Status epilepticus = seizure >5 minutes or recurrent seizures without return to baseline. The RAMPART trial (Silbergleit et al., NEJM 2012) proved IM midazolam is non-inferior to IV lorazepam, making it the ideal first-line agent when IV access is unavailable — a common rural scenario.

Tiered treatment protocol:

TIER 1: Emergent (0–5 min) — Benzodiazepines:

Route	Agent	Dose	Max
IV	Lorazepam	0.1 mg/kg	4 mg, repeat ×1
IM	Midazolam	0.2 mg/kg	10 mg
Intranasal	Midazolam	0.2 mg/kg	5 mg per nostril
Buccal	Midazolam	0.5 mg/kg	10 mg
Rectal	Diazepam	0.5 mg/kg	20 mg
IV	Diazepam	0.2–0.5 mg/kg	10 mg

TIER 2: Urgent (5–20 min) — Second-line agents:

Agent	Dose	Max	Notes
Levetiracetam	40–60 mg/kg IV over 5–15 min	4,500 mg	Fewest cardiovascular side effects; ESETT trial supports efficacy
Fosphenytoin	20 mg PE/kg IV (rate ≤3 mg PE/kg/min)	Add 5–10 PE/kg if needed	Cardiac monitoring required
Valproic acid	20–40 mg/kg IV over 5–10 min	3,000 mg	Avoid in age <2 yr , hepatic disease, suspected metabolic disorder
Phenobarbital	20 mg/kg IV (rate ≤1 mg/kg/min)	Add 5–10 mg/kg	Respiratory depression risk; preferred in neonates

TIER 3: Refractory (>30 min) — requires ICU transfer:

- Midazolam infusion: 0.2 mg/kg bolus then 0.05–0.4 mg/kg/hr
- Pentobarbital: 5–15 mg/kg load then 0.5–5 mg/kg/hr
- Ketamine: 1–3 mg/kg bolus then 0.5–3 mg/kg/hr (emerging evidence, complementary NMDA mechanism)

Febrile seizures: Simple (generalized, <15 min, single in 24 hr, age 6 mo–5 yr) requires no workup in a well-appearing child with identified source. Complex (>15 min, focal, recurrent) warrants further investigation including possible LP.

Febrile neonates under 60 days: the AAP 2021 guideline stratifies by age

This is one of the highest-stakes decisions in pediatric emergency medicine. The **AAP 2021 Clinical Practice Guideline** (Pantell et al., Pediatrics 2021) replaced the Rochester/Philadelphia/Boston criteria with a unified age-based approach for well-appearing, term infants 8–60 days with temperature $\geq 38.0^{\circ}\text{C}$.

Key inflammatory markers: Procalcitonin >0.5 ng/mL, CRP >20 mg/L, ANC >4,000/mm³. **WBC count is NOT recommended** for risk stratification.

8–21 days: Full sepsis workup mandatory — blood culture, UA/urine culture, LP with CSF studies. **Empiric antibiotics (ampicillin + gentamicin) and hospitalization for ALL infants.** Inflammatory markers do not change management at this age.

22–28 days: Obtain UA, blood culture, inflammatory markers. If **any marker is abnormal or UA positive** → **LP + hospitalize + empiric antibiotics**. If all normal and UA negative, LP is still recommended; if CSF normal, may consider discharge with close 24-hour follow-up (shared decision-making).

29–60 days (most significant departure from prior practice): Obtain UA, blood culture, inflammatory markers. If **UA negative AND all inflammatory markers normal** → **LP is NOT required; may discharge home without antibiotics** with reliable 24–36 hour follow-up. If any marker abnormal → LP + hospitalize + empiric antibiotics. For >28 days, **ceftriaxone** is appropriate (contraindicated in neonates <28 days due to bilirubin displacement risk).

Add acyclovir 20 mg/kg IV q8h if HSV is suspected: vesicular rash, seizures, CSF pleocytosis, elevated LFTs, maternal HSV history, or failure to improve on antibiotics.

Neonatal CSF normal values differ markedly from older children: WBC <10–22 cells/ μL , protein up to 110–115 mg/dL (neonates), glucose $>\frac{2}{3}$ of serum glucose. Neonatal meningitis can occur with normal CSF in up to 12% of cases — always send culture.

PART VIII: BURNS, FRACTURES, AND SURGICAL EMERGENCIES

Pediatric burns require modified Parkland formula plus maintenance fluids

TBSA calculation uses the **modified Rule of 9s** (infant head = 18% vs. 9% in adults; each leg = 14% vs. 18%). The **Lund-Browder chart** is more accurate and preferred.

Fluid resuscitation: $3\text{--}4 \text{ mL} \times \text{kg} \times \% \text{ TBSA burned}$ of Lactated Ringer's over 24 hours. Give **half in the first 8 hours** (from time of burn), remaining half over the next 16. **Critically, pediatric patients also need maintenance fluids (Holliday-Segar) added on top.** For children <20 kg, use **D5LR for maintenance** to prevent hypoglycemia. Fluid initiation threshold is >10% TBSA in children (vs. >20% in adults).

Monitor urine output: **1 mL/kg/hr for children <30 kg**, 0.5–1 mL/kg/hr for ≥30 kg. For inhalation injury (enclosed space, singed nasal hairs, carbonaceous sputum), perform **early intubation** — pediatric airway edema develops rapidly.

Transfer to burn center for: >10% TBSA partial thickness, any full thickness, face/hands/feet/perineum/joint involvement, electrical/chemical burns, inhalation injury.

Salter-Harris fractures and compartment syndrome recognition

Salter-Harris classification (SALTR mnemonic):

Type	Description	Frequency	Management
I	Through physis only (often normal X-ray)	6%	Splint + ortho follow-up
II	Through physis + metaphysis (Thurston-Holland fragment)	75% (most common)	Closed reduction + splint
III	Through physis + epiphysis (intra-articular)	8%	Often requires ORIF
IV	Through metaphysis + physis + epiphysis	10%	Requires ORIF
V	Crush injury to physis	<1%	Often diagnosed retrospectively

Compartment syndrome in children is recognized by **the "3 A's"** (more reliable than classic 6 P's in children): **Anxiety, Agitation, and increasing Analgesic requirements.** Pain out of proportion to injury, especially with passive stretch, is the most reliable early sign. Pulselessness and paralysis are **late, ominous findings. Remove all splints/casts immediately** and measure compartment pressure if available (delta pressure ≤30 mmHg of diastolic = fasciotomy indicated). **Fasciotomy must occur within 6–8 hours** to prevent irreversible ischemic damage.

Three abdominal emergencies that cannot wait

Appendicitis — Pediatric Appendicitis Score (PAS): Migration of pain (1), anorexia (1), nausea/vomiting (1), RLQ tenderness (2), cough/hop/percussion tenderness (2), fever >38°C (1), leukocytosis >10,000 (1),

neutrophilia $>7,500$ (1). Total: 10 points. Score ≤ 3 = low risk (consider discharge); 4–6 = imaging (ultrasound first); ≥ 7 = high risk (surgical consultation). Antibiotics: ceftriaxone + metronidazole.

Intussusception: Peak age 6–36 months. Classic triad (colicky pain, vomiting, currant jelly stools) is present in only 20–40%. Ultrasound shows **target/donut sign**. Treatment is air or contrast enema reduction (80–95% success) — unavailable at most rural EDs, so **stabilize (IV access, fluids, NPO) and transfer**.

Malrotation with midgut volvulus: Bilious vomiting in a neonate is a surgical emergency until proven otherwise. Can cause complete midgut necrosis within hours. Upper GI series is gold standard (corkscrew appearance). Rural ED management: NPO, NG decompression, aggressive IV fluid resuscitation, broad-spectrum antibiotics (ampicillin + gentamicin + metronidazole), and **emergent transfer for Ladd procedure**.

PART IX: CARDIAC EMERGENCIES, CHILD ABUSE, AND NEONATAL EMERGENCIES

SVT in infants presents as heart failure — adenosine technique matters

SVT recognition: narrow complex tachycardia with rates typically **>220 bpm in infants, >180 bpm in children**. Try vagal maneuvers first: ice bag to upper face (infants), Valsalva (older children).

Adenosine: 0.1 mg/kg rapid IV push (max 6 mg first dose), then 0.2 mg/kg (max 12 mg second dose). **Must be given as rapid push** through stopcock closest to patient followed by immediate 5–10 mL NS flush. If hemodynamically unstable: **synchronized cardioversion 0.5–1 J/kg**, increase to 2 J/kg.

PALS cardiac arrest key doses: Epinephrine 0.01 mg/kg IV/IO (1:10,000) every 3–5 minutes. Defibrillation for VF/pVT: 2 J/kg $\rightarrow$ 4 J/kg $\rightarrow$ 4–10 J/kg. Amiodarone 5 mg/kg IV (max 300 mg) or lidocaine 1 mg/kg for refractory shockable rhythms. Compression-to-ventilation ratio: **15:2** (two rescuers).

Child abuse recognition and mandatory reporting

Sentinel injuries are the most critical early finding: **any bruise in a pre-mobile infant** ("Those who don't cruise rarely bruise"), frenulum tears, and subconjunctival hemorrhages in non-neonatal infants. **25% of children diagnosed with abuse had a prior sentinel injury that was missed.**

High-specificity fracture patterns for abuse: Classic metaphyseal lesions (bucket-handle/corner fractures), **posterior rib fractures**, sternal fractures, and scapular fractures. **Spiral fractures are no longer considered pathognomonic** — a common and dangerous misconception.

Required actions: Skeletal survey in all children <2 years with suspected abuse. Head CT in all children <6 months with suspected abuse. Document objectively with body diagrams and photographs with scale markers. **Mandatory reporting in all 50 states** based on reasonable suspicion, not certainty. Dilated fundoscopic exam for suspected abusive head trauma (retinal hemorrhages present in 60–85%).

Neonatal emergencies: resuscitation, congenital heart disease, and hyperbilirubinemia

NRP algorithm: Warm, dry, stimulate, position, suction if needed. If apneic/gasping or HR <100 → PPV at 40–60 breaths/min. If HR <60 after 30 seconds of effective PPV → chest compressions (3:1 ratio). If HR still <60 → epinephrine 0.01–0.03 mg/kg IV/UVC (or 0.05–0.1 mg/kg ET).

Congenital heart disease presents at 1–2 weeks as the ductus closes. Two presentations: **cyanotic** (ductal-dependent pulmonary flow — TGA, pulmonary atresia, tricuspid atresia) and **shock** (ductal-dependent systemic flow — coarctation, HLHS, interrupted aortic arch). **Hyperoxia test:** PaO₂ <100 mmHg on 100% FiO₂ strongly suggests cardiac etiology. Start PGE1, prepare for intubation, and transfer emergently.

Neonatal hyperbilirubinemia (AAP 2022 guideline): Use hour-specific TSB thresholds stratified by gestational age and neurotoxicity risk factors (albumin <3.0, isoimmune hemolytic disease, G6PD deficiency, sepsis). Signs of **acute bilirubin encephalopathy** (lethargy, high-pitched cry, arching/opisthotonus) demand **immediate intensive phototherapy and preparation for exchange transfusion** regardless of TSB level. **IVIG 0.5–1 g/kg** may be given for isoimmune disease approaching exchange threshold.

PART X: PAIN MANAGEMENT, PROCEDURAL SEDATION, AND TRANSFER

Pain management should be multimodal and weight-based

Severity	Agent	Dose	Route
Mild	Acetaminophen	15 mg/kg q4–6h (max 75 mg/kg/day)	PO/PR
Mild	Ibuprofen	10 mg/kg q6–8h (max 40 mg/kg/day)	PO
Moderate	Intranasal fentanyl	1.5–2 mcg/kg (max 100 mcg)	IN
Moderate	Ketorolac	0.5 mg/kg IV (max 15 mg <50 kg)	IV
Severe	Morphine	0.1 mg/kg q3–4h (max 5 mg)	IV
Severe	Fentanyl	1–2 mcg/kg	IV

Intranasal fentanyl is a rural ED game-changer — rapid onset without IV access. Use FLACC scale (0–10) for preverbal children, FACES scale for ages 4–12, and numeric rating scale for older children.

Procedural sedation with ketamine is the rural workhorse

Ketamine: 1–2 mg/kg IV or 4–5 mg/kg IM. Onset: 1 min IV, 5 min IM. Duration: 10–20 min IV, 20–40 min IM. Emergence reactions are less common in children than adults. Maintain NPO when feasible (2 hr clear liquids, 6 hr solids), but **emergencies take priority over fasting status**. Monitoring: continuous pulse oximetry, ETCO₂, cardiac monitoring, BP q5 min, and a dedicated observer.

SBAR framework for transfer communication

Element	Content
S — Situation	Patient age, weight, chief complaint, current status, interventions performed
B — Background	Medical history, allergies, medications, events leading to presentation
A — Assessment	Current vital signs, physical exam findings, lab/imaging results, working diagnosis, severity classification
R — Recommendation	Specific needs (PICU, surgery, specialty), urgency of transfer, mode of transport, interventions needed en route

Pre-transfer stabilization checklist:

- Airway secured or plan in place
- IV/IO access established (minimum 2 for critically ill)
- Weight-based medication doses calculated and documented
- Copies of all labs, imaging, and clinical notes prepared
- Current infusions labeled with concentrations and rates
- Parents informed and consent for transfer obtained
- EMTALA requirements satisfied (accepting physician identified, risks/benefits of transfer documented)

Telemedicine transforms rural pediatric emergency care

Telemedicine programs (such as REACH — Remote Emergency Access for Children's Health) provide real-time video consultation with pediatric emergency physicians and subspecialists. Evidence shows telemedicine in rural EDs **reduces unnecessary transfers, improves diagnostic accuracy, and increases provider confidence**. Conditions that benefit most: sepsis (real-time resuscitation coaching), DKA management, RSI guidance, cardiac arrhythmia interpretation, child abuse evaluation, and neonatal resuscitation. Programs like **Project ECHO** provide ongoing case-based education to maintain competency between critical events.

PART XI: EQUIPMENT AND MEDICATION READINESS

Every rural ED needs a pediatric-ready resuscitation bay

The AAP/ACEP Joint Policy Statement on Emergency Department Equipment for Pediatric Patients specifies essential items by category:

Airway: ETT sizes 2.5–8.0 (cuffed and uncuffed), laryngoscope blades (Miller 0, 1, 2; Mac 2, 3), LMA sizes 1–4, BVM sizes (infant, child, adult), suction catheters (6–14 Fr), nasopharyngeal and oropharyngeal airways in pediatric sizes, end-tidal CO₂ detectors

Vascular access: IO needles (EZ-IO with pediatric and adult needles — **the single most important backup access device**), pediatric IV catheters (24–18 gauge), umbilical catheterization supplies, pediatric blood pressure cuffs

Monitoring: Pulse oximetry with neonatal/pediatric probes, ETCO₂ capable monitoring, cardiac monitors with pediatric electrode options

Medications that must be stocked: Epinephrine (1:1,000 and 1:10,000), atropine, adenosine, amiodarone, dextrose (D10W, D25W), rocuronium, succinylcholine, ketamine, midazolam, lorazepam, fosphenytoin/levetiracetam, albuterol, racemic epinephrine, dexamethasone, ceftriaxone, ampicillin, gentamicin, acyclovir, mannitol, 3% hypertonic saline, prostaglandin E1, ondansetron

Cognitive aids: Broselow tape or equivalent length-based system, laminated pediatric DKA protocol, RSI checklist, sepsis bundle checklist, and transfer communication template.

Conclusion: five principles that save pediatric lives in rural emergency departments

This guide covers an enormous breadth of pediatric emergencies, but the principles that cut across every condition are strikingly consistent. **First, recognize severity early** — children compensate brilliantly until they cannot, so tachycardia and subtle perfusion changes matter more than blood pressure. **Second, weight-based precision is non-negotiable** — a 10-fold dosing error in a 3 kg neonate can be lethal, making Broselow tape and cognitive aids lifesaving tools rather than crutches. **Third, certain diagnoses demand immediate pattern recognition** — bilious vomiting in a neonate (volvulus), bruising in a pre-mobile infant (abuse), and cyanosis unresponsive to oxygen (ductal-dependent heart disease) require reflexive action, not deliberation. **Fourth, transfer early and communicate well** — SBAR-formatted calls, telemedicine consultation, and early transport activation prevent the most dangerous rural scenario: a deteriorating child hours from definitive care. **Fifth, practice before the crisis arrives** — quarterly simulation drills, pre-calculated dosing sheets, and equipment checks convert rare events from chaos into managed complexity. The evidence synthesized here represents guidelines current through early 2026, but pediatric emergency medicine evolves rapidly; annual review of AAP, ACEP, and AHA/PALS updates is essential to maintain the standard of care these children deserve.